

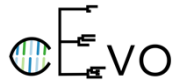
BRIDGING STRATIGRAPHIC PALAEOBIOLOGY AND PHYLOGENETICS

COMMUNITY DRIVEN PHYLOGENETIC SOFTWARE DEVELOPMENT

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1 October 2026



ETH zürich



COMMUNITY BAYESIAN PHYLOGENETIC SOFTWARE

MrBayes, BEAST X

BUILT-IN MODEL COMPONENTS

Users combine built-in model components.

A new model is added by the core team and ships with the next release.

RevBayes

USERS COMPOSE MODELS

Users build new models from existing distributions, functions and moves in the Rev language.

A new component goes into the core: reviewed and consistent, but waiting on the core team.

BEAST 2

INDEPENDENT PACKAGES DEFINE NEW COMPONENTS/MODELS

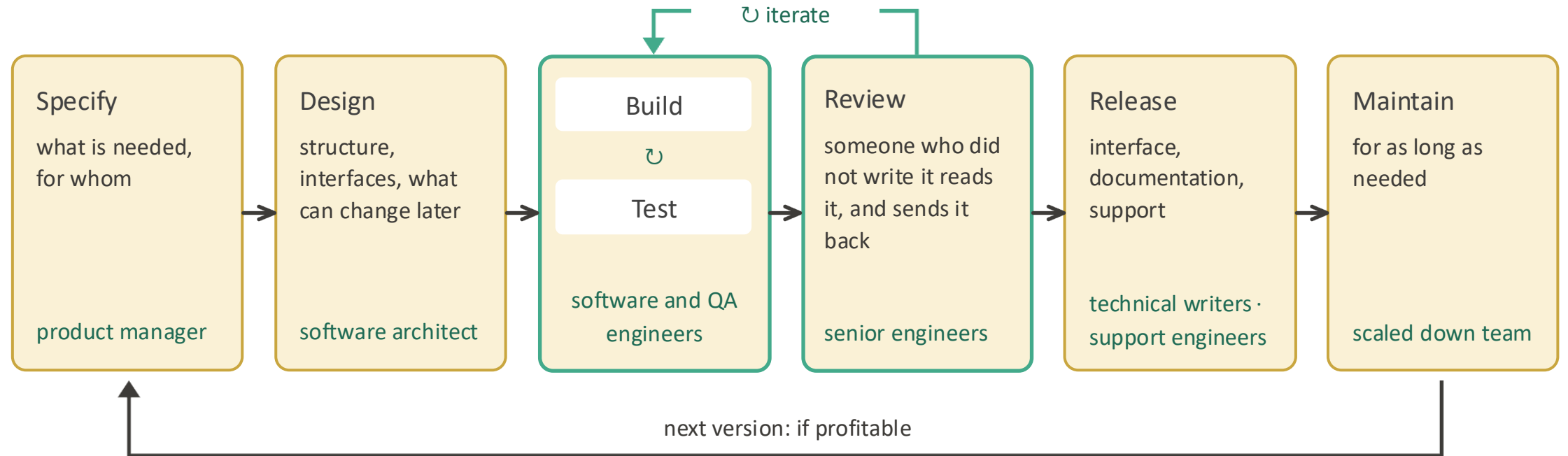
The core is an MCMC engine adapted for phylogenetics. A package can add almost anything to it: new models, tree/network types, operators, GUI panels.

PART 1

IMPLEMENTING AND VALIDATING A NEW MODEL

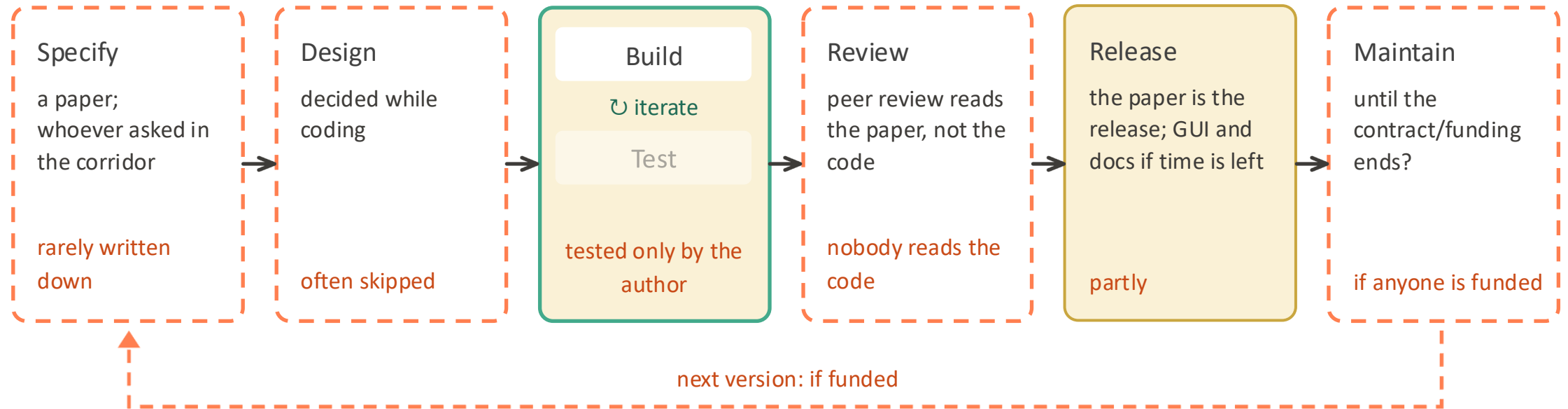
HOW SOFTWARE GETS BUILT

IN INDUSTRY: A CYCLE, WITH ROLES



HOW SOFTWARE GETS BUILT

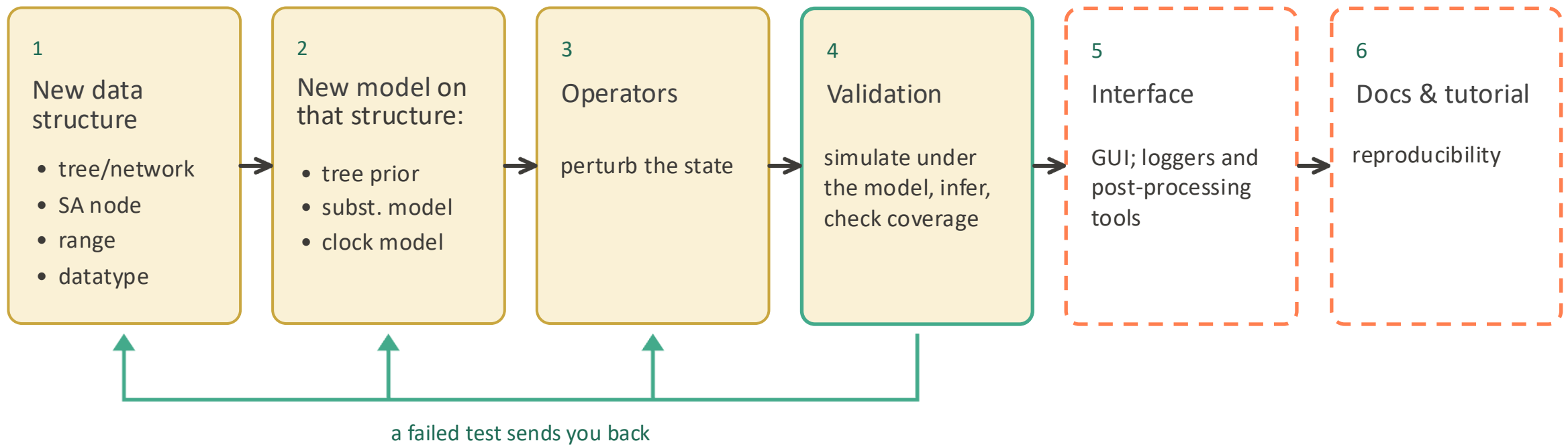
IN SCIENCE: THE SAME CYCLE, ONE PERSON



- Every role is the same PhD/postdoc, part-time, with a deadline.
- Software is a means to an end not the goal itself!

HOW SOFTWARE GETS BUILT => BUILD AND TEST

WHAT IMPLEMENTING A BAYESIAN PHYLOGENETIC MODEL INVOLVES



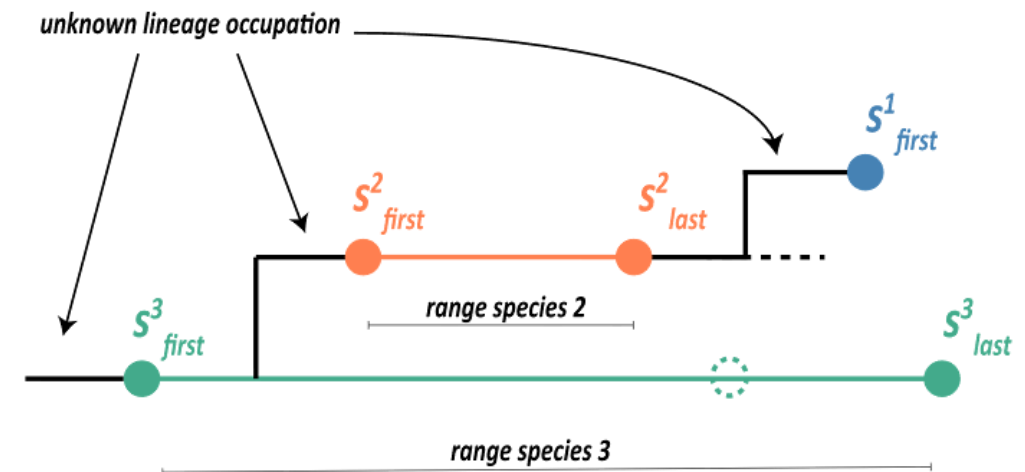
1 DATA STRUCTURE

COMPLEX OBJECTS

- Extending a basic model often needs a more complex data structures.
- Everything downstream must adopt it: operators, loggers, summary and plotting tools.

In sRanges

A taxon occupies a stretch of an oriented tree branch. No BEAST 2 tree could hold that.



A DEFINITION IS NOT AN ALGORITHM

The maths says what to compute but code decides how.

- How to traverse the tree?
- How to store and restore the state?
- What to cache and when to recompute?

In `sRanges`

The tree carries orientation: which child is the ancestor. Storing and restoring the tree state had to change to carry it too.

Otherwise a rejected move could leave the right topology with the wrong orientation.

3 OPERATORS

THE SAMPLER MUST LEARN THE NEW DATA STRUCTURE

- Moves must be able explore the full space.
- Constraints break existing moves.
- Each new move must be reversible, with the Hastings correction derived.
- **Bugs are silent on analysis:** the chain converges anyway.

In sRanges

Standard moves cut ranges apart, leaving a taxon in two places or an ancestor after its descendant.

ENSURING CORRECTNESS

- Unit tests and code review.
- Sampling vs simulating under the prior.
- **Coverage**: draw from the prior, simulate, infer. The true value falls inside the X% HPD interval X% of the time.
- Good coverage != inference power. Bias, error, relative hpd width are measured separately.

Mendes, et al.2025

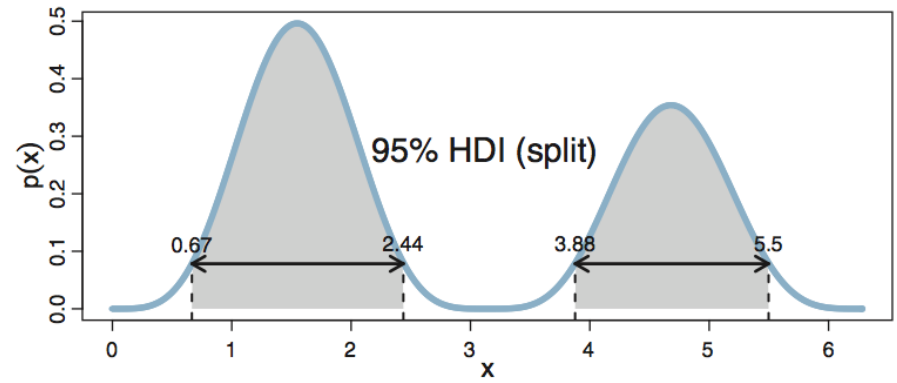
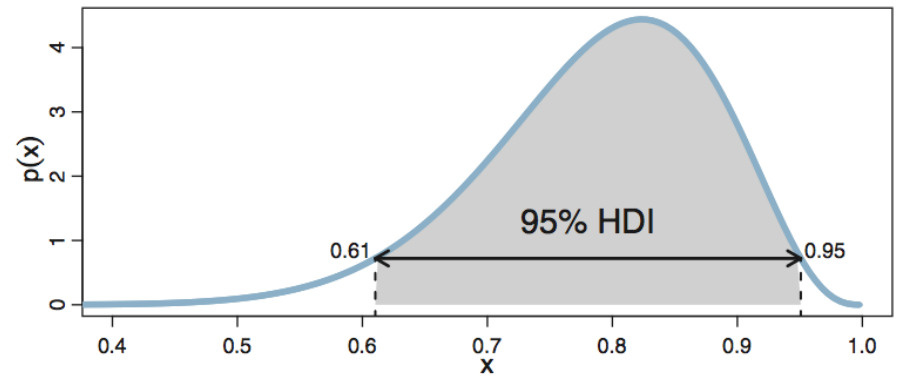
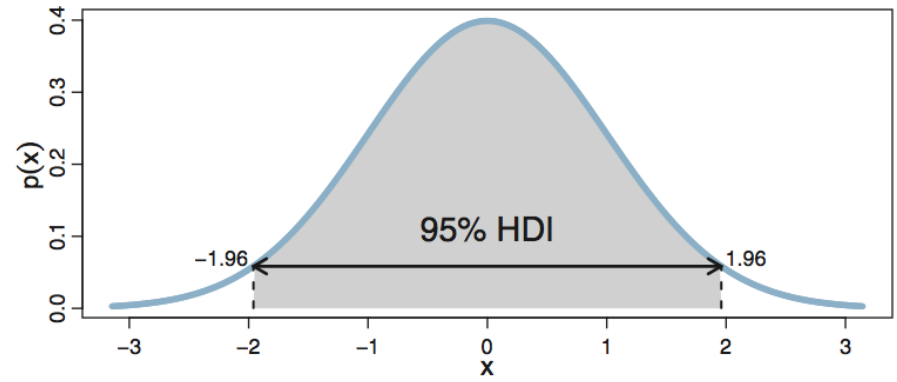
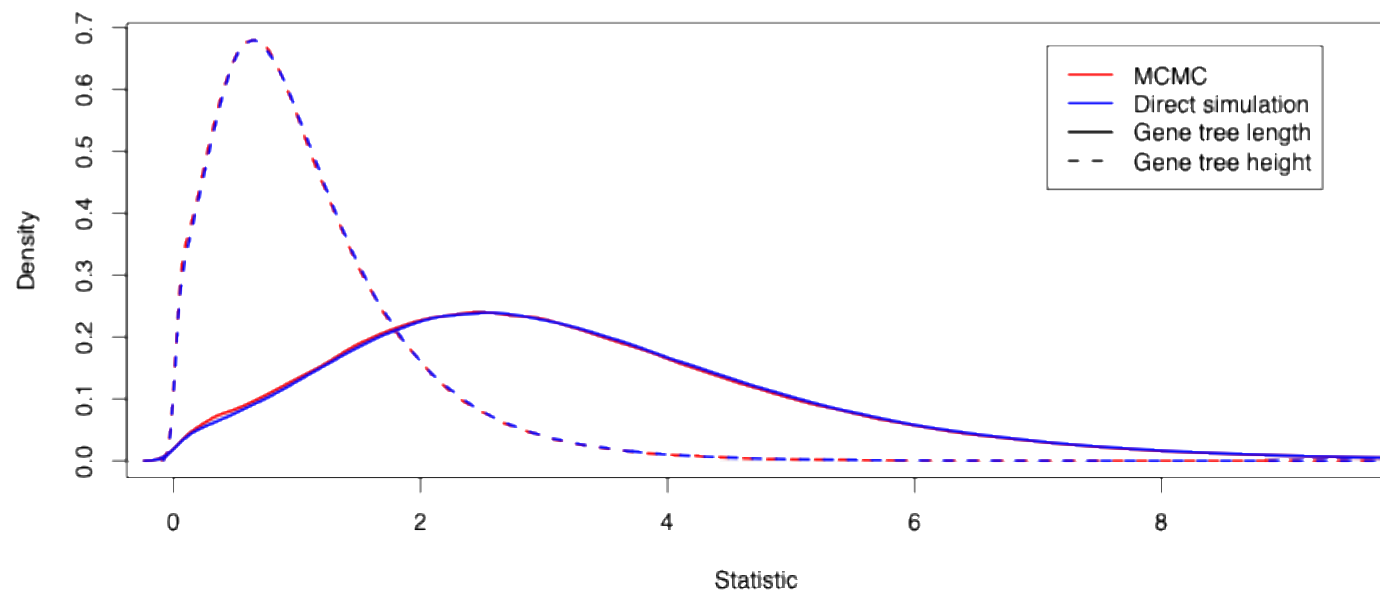
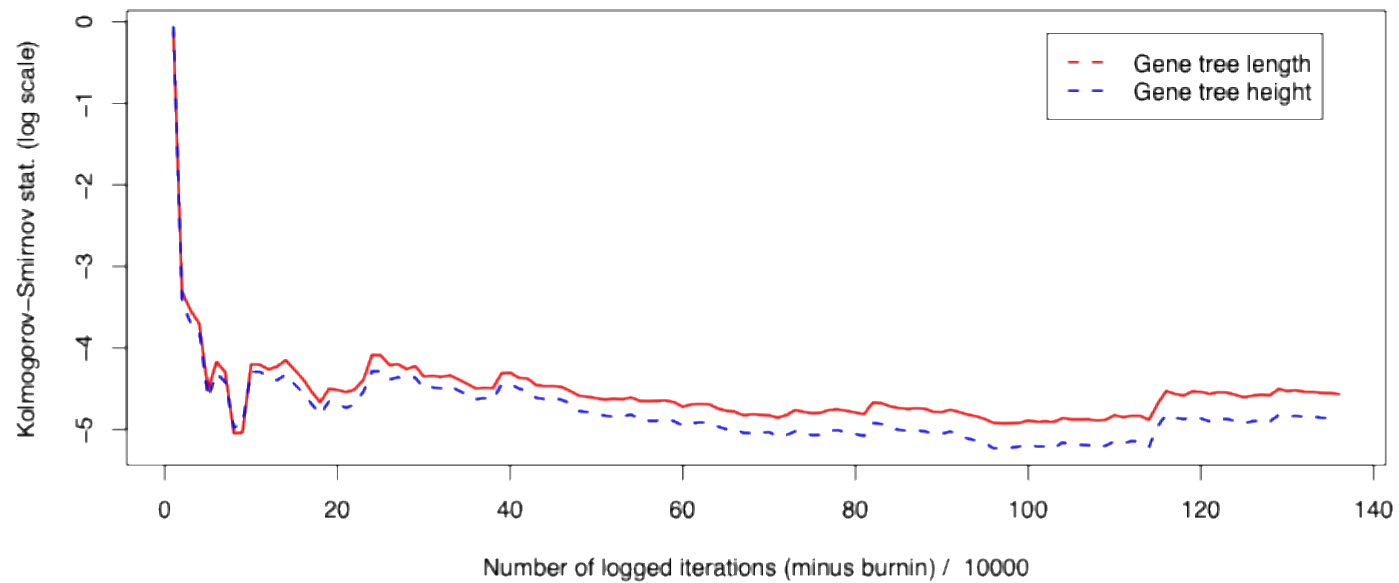


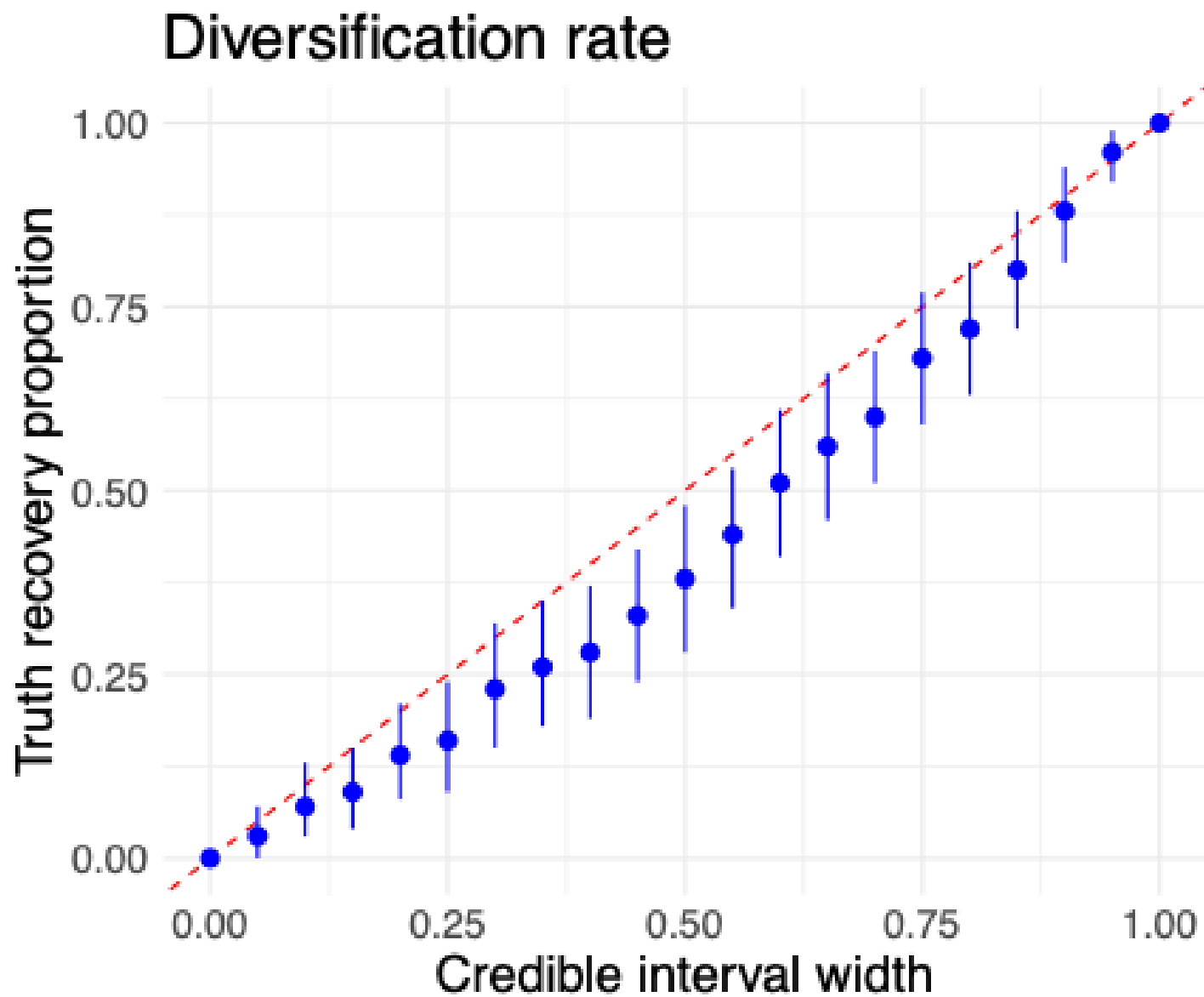
Figure 4.5 Examples of 95% highest density intervals (HDIs). For each example, all the x values inside the interval have higher density than any x value outside the interval, and the total mass of the points inside the interval is 95%. The 95% area is shaded, and it includes the zone below the horizontal arrow. The horizontal arrow indicates the width of the 95% HDI, with its ends annotated by (rounded) x values. The height of the horizontal arrow marks the minimal density exceeded by all x values inside the 95% HDI.



Kolmogorov–Smirnov test for network statistics



Check multiple
interval widths!



YOU CAN SIMULATE FAR MORE THAN YOU CAN INFER

- Inference takes days, so the study must stay small.
- Discarding oversized simulations biases the test.
 - Difficult to provide the same conditioning!
- Only affordable regimes get covered.

In sRanges

A birth–death process from a broad prior gives broad range of tree sizes.

We searched for regimes that give workable trees, minimal rejection. Ideally, model and simulation would be conditioned the same way.

AFTER VALIDATION

CORRECT IS NOT THE SAME AS INFORMATIVE

- Contraction from (implied!) prior to posterior.
- Needs real(-istic) data and the people who understand them.
- Here the developer may not be the right person to lead.

IS USABILITY A REQUIREMENT?

- GUI, loggers, summary tools, documentation, tutorial.
- None of it is required to publish, so it gets cut.
- Post-processing is also modelling decision.
- Without a tutorial, maybe three people can run it?

WHAT IS SPECIFIC TO PHYLOGENETICS

The tree is the parameter

Deriving the tree prior is half the maths. Exploring tree space needs operators with their own derivations, and topological restrictions such as ranges make both harder.

A new model often needs new data structure

That implies new caching, operators, loggers, post-processing.

Simulation outruns inference

We can simulate far larger and more complex scenarios than we can infer. This limits validation.

No ground truth, slow feedback

Real data often cannot show the code is right, only that results are plausible. One validation run is hours to days of MCMC, so a test that takes a week is not run.

GENERAL PROBLEM

In scientific software, one person is

- lead dev,
 - junior dev,
 - tester,
 - GUI dev,
 - documentation writer,
 - teacher.
-
- All part-time (in addition to research!), on a limited contract. Under pressure to get results.

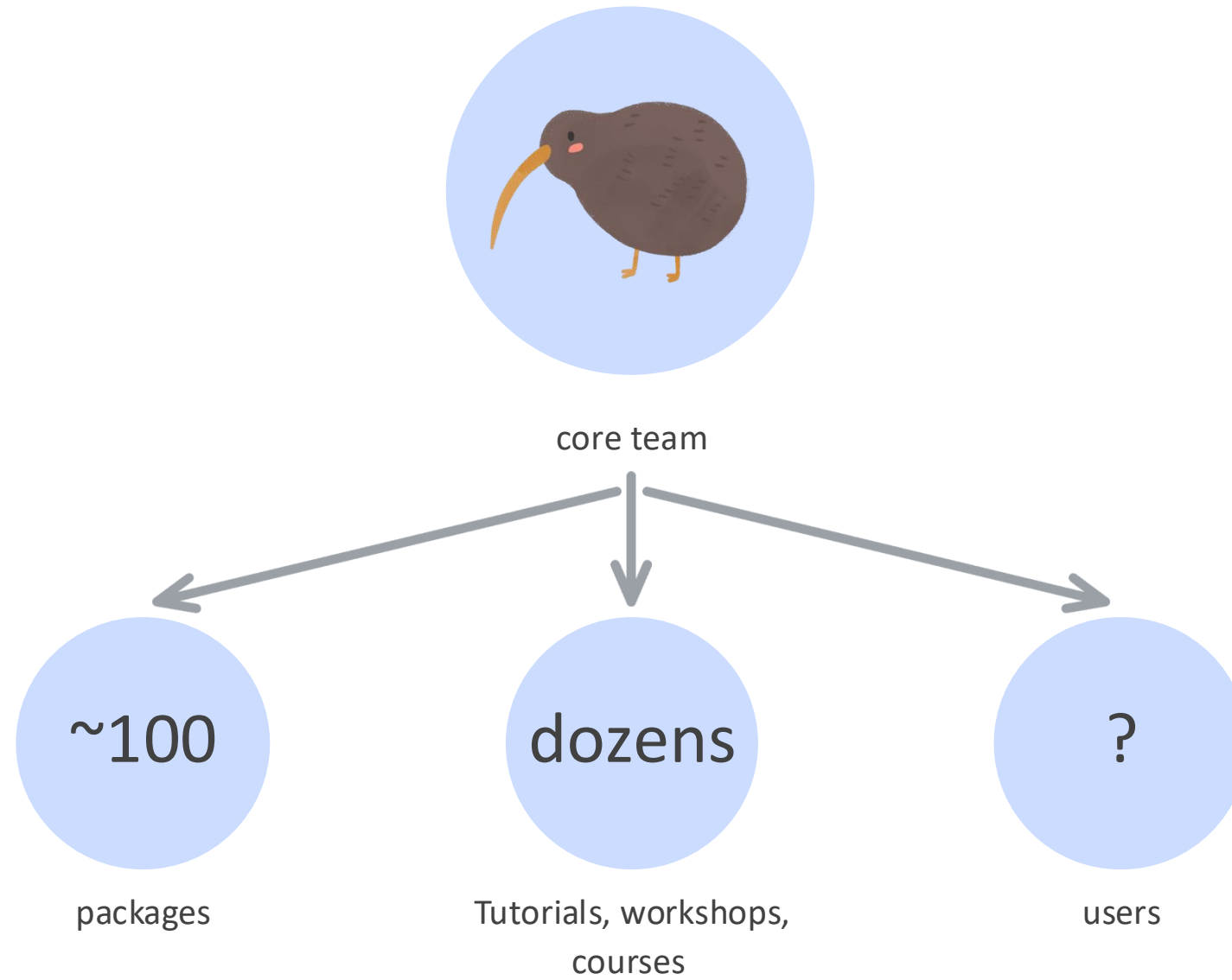
THE MISSING ROLE?

Research software engineer. The Software Sustainability Institute (UK), deRSE (Germany) and the Netherlands eScience Center
It can be a career rather than a phase.

PART 2

WHY THE COMMUNITY MATTERS FOR A BIG SOFTWARE PROJECT

BEAST 2



WHAT HAPPENED WITHOUT COORDINATION

FOR DEVELOPERS

- Core updates arrived as a surprise: welcome but blindsiding.
- Keeping a package working meant dropping whatever else the owner was doing.
- Someone picked up the slack, or nobody did and the package/tutorial stopped working.
- No one has good overview: risk of removing too much or dragging outdated stuff along.

FOR USERS

- Outdated tutorials.
- Package never moved => forced to use old release.
- Useful functionality removed.
- Old analyses not reproducible.
- Opportunity to request new features missed.

THE STAKEHOLDER MEETINGS

- No more surprises: everyone learns about a core change before it lands, with time to react.
- Decisions about the platform taken as a community.
- Decisions documented: what does/does not change, when, who does what.
- Nobody is anyone's manager, so alignment is voluntary.

SINCE SEPTEMBER 2025

Every 6–8 weeks. Core team, package developers, workshop organisers, power users.

Standing agenda: release status, package migration, tutorials, workshops, forum, new packages.

NOTHING NEW

What the stakeholder group is doing	What it is called	How others do it
Deprecate first, remove a release later	Deprecation policy; semantic versioning	NumPy NEP 23; semver.org
Core changes discussed before they land	Steering council instead of a single decider	Python PEP 13; Rust RFC process
Ways forward documented, not just agreed in a call	Request for comments (RFC)	IETF, Rust, React
A package whose owner has moved on	Bus factor; named maintainers	Linux kernel MAINTAINERS file
Tutorials that match the release you run	Versioned documentation; long-term-support releases	Ubuntu LTS; readthedocs versions

TAMING THE BEAST: TEACHING AS SHARED INFRASTRUCTURE

- Community-written tutorials and week-long workshops.
- Every core release means re-running every XML, re-taking every screenshot, re-reading every sentence. Hours per tutorial, dozens of tutorials.

WHY IT IS ON THE STAKEHOLDER AGENDA

Who updates the tutorials, and by when, must be decided with the release. Otherwise the release ships and the teaching material stops matching it.

BEAST 2.7

Troubleshooting initialization issues

Getting BEAST2 to start

Joëlle Barido-Sottani, Louis du Plessis

🕒 Updated Jun 2026

Troubleshooting SA ORC

BEAST 2.7

Troubleshooting-convergence-issues

Post-processing and improving performance

David A. Rasmussen

🕒 Updated Jun 2026

Troubleshooting

BEAST 2.7

Prior-selection

Prior selection and clock calibration using Influenza A data.

Veronika Bošková, Venelin Mitov, Louis du Plessis

🕒 Updated Jun 2026

Basic workflow Epidemiology BDSKY

BEAST 2.7

Structured birth-death model with BDMM-Prime

Population structure using the multi-type birth-death model

Denise Kühnert, Jūlija Pečerska, Ugnė Stolz, Tim Vaughan

🕒 Updated May 2026

BEAST 2.7

Species Trees and Species Delimitation with SNAPPER

Analyzing SNP or AFLP data

Adam Leaché, Remco Bouckaert

🕒 Updated Nov 2025

Advanced analysis Population genetics Macroevolution snapper

BEAST 2.7

Total Evidence Tutorial

Inferring phylogenies using genetic sequences and morphological characters

Bethany J. Allen, Joëlle Barido-Sottani

🕒 Updated Aug 2025

Advanced analysis Macroevolution Palaeontology MM SA

ONE YEAR IN, HONESTLY

NOT YET WORKING AS INTENDED

- Little communication in-between meetings.
- Both sides still sometimes struggle to hear each other.
- Heavily lead by the core team, others take the backseat.

WHAT IT HAS CHANGED

- Decisions are clearly formulated, documented and explained.
- Problems surfaced and were fixed by the core team before release.
- Deprecate first, remove later policy. A clear signal but also time for people who need to run analyses, not migrate code.
- Information flows in both directions.

FROM A SUPPORT LIST TO A PLACE TO TALK

The screenshot shows the GitHub interface for the `CompEvol / beast2` repository. The top navigation bar includes links for Code, Issues (200), Pull requests, Agents, Actions, Security and quality, and Insights. A search bar is located on the right. The left sidebar contains filters for Issues (Assigned to me, Created by me, Mentioned, Recent activity) and Views (Views, Milestones, Labels). The main content area is titled 'Issues' and features a search bar with the query `is:issue state:open`. Below the search bar, there are filters for Open (200) and Closed (924) issues, along with dropdown menus for Author, Labels, Projects, Milestones, Assignees, and Types. A 'New issue' button is in the top right corner. The list of issues includes:

- How should equilibrium frequencies be handled in a GeneralSubstitutionModel with a potentially non-time-reversible rate matrix?** (1 comment) - CompEvol/beast2#1217 · jjmccollum opened 4d ago
- For GeneralSubstitutionModel, use Padé approximant if matrix is non-diagonalizable or ill-conditioned** (2 comments) - CompEvol/beast2#1216 · jjmccollum opened 5d ago
- Bundled Java runtime in BEAST 2.7.x releases has known vulnerabilities** (2 comments) - CompEvol/beast2#1215 · berkleycam opened on Jun 16
- EpochFlexOperator causes infinite loop/livelock when applied to 2-trees**

FROM A SUPPORT LIST TO A PLACE TO TALK

The screenshot displays the BEAST Community Forum interface. At the top, there are navigation links for BEAST X Docs, BEAST X Source, BEAST 2 Docs, and BEAST 2 Source. A search icon, a chat icon, and a user profile icon are also present. Below the navigation bar, there are tabs for 'categories >', 'tags >', 'Latest', 'New', 'Hot', and 'Categories' (which is selected). A 'New Topic' button is located on the right. The main content area is divided into two columns. The left column lists categories with their respective topic counts: 'Announcements' (2), 'Help & Support' (0), 'Packages & Extensions' (1), and 'Development' (0). Each category has a brief description and a list of sub-topics. The right column shows a list of topics, each with a user icon, a title, a category tag, a vote count, and a date. The topics listed are: 'BEAST X — Tutorial Index', 'Taming the BEAST — Tutorial Index', 'BEAST 2.7.8 Released', 'BEAST X v10.5.0 Released', 'How to use this forum', and 'BEAST2 package directories'.

Category	Topics	Latest
Announcements Software releases, workshop and conference notices, and community news.	2	BEAST X — Tutorial Index Tutorials & Workflows getting-started ■ beast-x 0 votes Feb 19
Help & Support Ask questions about using BEAST X and BEAST 2. Search first — your question may already be answered. ■ Installation & Setup ■ Model Specification & Priors ■ Running Analyses ■ Results & Interpretation	0	Taming the BEAST — Tutorial Index Tutorials & Workflows beast2 ■ getting-started 0 votes Feb 19
Packages & Extensions Discussion of BEAST 2 packages and BEAST X extensions.	1	BEAST 2.7.8 Released Announcements beast2 ■ release ■ v2-7-8 Feb 19
Development For contributors and developers: feature proposals, architecture, build and CI discussion. ■ BEAST X Development ■ BEAST 2 Development ■ Package Development	0	BEAST X v10.5.0 Released Announcements release ■ beast-x ■ v10-5 Feb 19
		How to use this forum Site Info Feb 19
		BEAST2 package directories Packages & Extensions ■ beast2 0 votes Feb 19

WHAT CAN YOU DO TODAY

Say what you need before the code exists

Which data you have, how it is coded, what question you want answered.

Adopt early, report reproducibly

Public and open source from the beginning. Early adopters are the only independent testers a package gets before release.

Lead the application, share the credit

A developer implementing and validating a model, and an empiricist leading the study that uses it, with authorship reflecting the split.

Cite the software, credit the maintainers

Plan and name who maintains the software, interface, documentation and tutorials.

FOR THE DISCUSSION

TWO OPEN QUESTIONS

QUESTION 1

ONE WORKING GROUP, OR LOOSE COORDINATION?

	One group: regular meetings, shared deadlines	Loose coordination: independent groups, occasional alignment
For	Shared standards and review; duplication caught early; newcomers have somewhere to arrive; visible ownership.	Fits how academia is funded; fast for the individual; no meeting nobody has time for.
Against	PhDs and postdocs run on different clocks and under pressure to publish; deadlines compete with theses and papers; who leads when nobody is funded to?	Duplicated operators, incompatible packages, knowledge in heads; every update is a surprise.

HOW TO BEST SET IT UP AND RUN IT?

QUESTION 2

MANY EXTENSIONS: ONE PACKAGE, OR SEVERAL?

	Separate packages, one per variant	One package, shared core, application layers
For	Independence and speed; clear authorship and credit; a broken package breaks only itself.	One install for the user; consistent names and defaults; the shared core is tested once and better.
Against	Repetition of code and documentation; incompatibilities; N packages to migrate on every core change.	Who approves a change? Whose best practice wins? One maintainer can become the bottleneck. Overly complicated set up, steep learning curve.